



US 20250283301A1

(19) **United States**(12) **Patent Application Publication**
HASHIMOTO et al.(10) **Pub. No.: US 2025/0283301 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **CONTROL DEVICE FOR EXCAVATOR***E02F 9/22* (2006.01)*E02F 9/24* (2006.01)(71) Applicant: **SUMITOMO CONSTRUCTION**
MACHINERY CO., LTD., Tokyo (JP)(52) **U.S. Cl.***E02F 9/26* (2006.01)(72) Inventors: **Kazutoshi HASHIMOTO**, Chiba (JP);
Hiroyuki TSUKAMOTO, Chiba (JP)CPC *E02F 9/2041* (2013.01); *E02F 3/964*
(2013.01); *E02F 9/2025* (2013.01); *E02F 9/24*
(2013.01); *E02F 9/261* (2013.01); *E02F*
9/2203 (2013.01); *E02F 9/2225* (2013.01);
E02F 9/2285 (2013.01); *E02F 9/2292*
(2013.01); *E02F 9/2296* (2013.01)(21) Appl. No.: **19/215,875**(22) Filed: **May 22, 2025****Related U.S. Application Data**(63) Continuation of application No. PCT/JP2023/
040560, filed on Nov. 10, 2023.**Foreign Application Priority Data**

Nov. 24, 2022 (JP) 2022-187326

Publication Classification(51) **Int. Cl.***E02F 9/20* (2006.01)*E02F 3/96* (2006.01)

(57)

ABSTRACT

A control device for an excavator is disclosed. The excavator includes an upper slewing body, an attachment provided on the upper slewing body, a slewing mechanism, a lower traveling body, and a blade provided on the lower traveling body. The control device includes circuitry configured to receive an input of an operation command for instructing an operation of the attachment, determine whether a person is detected around the excavator, and cause the excavator to perform an operation of lowering the blade when the person is not detected.

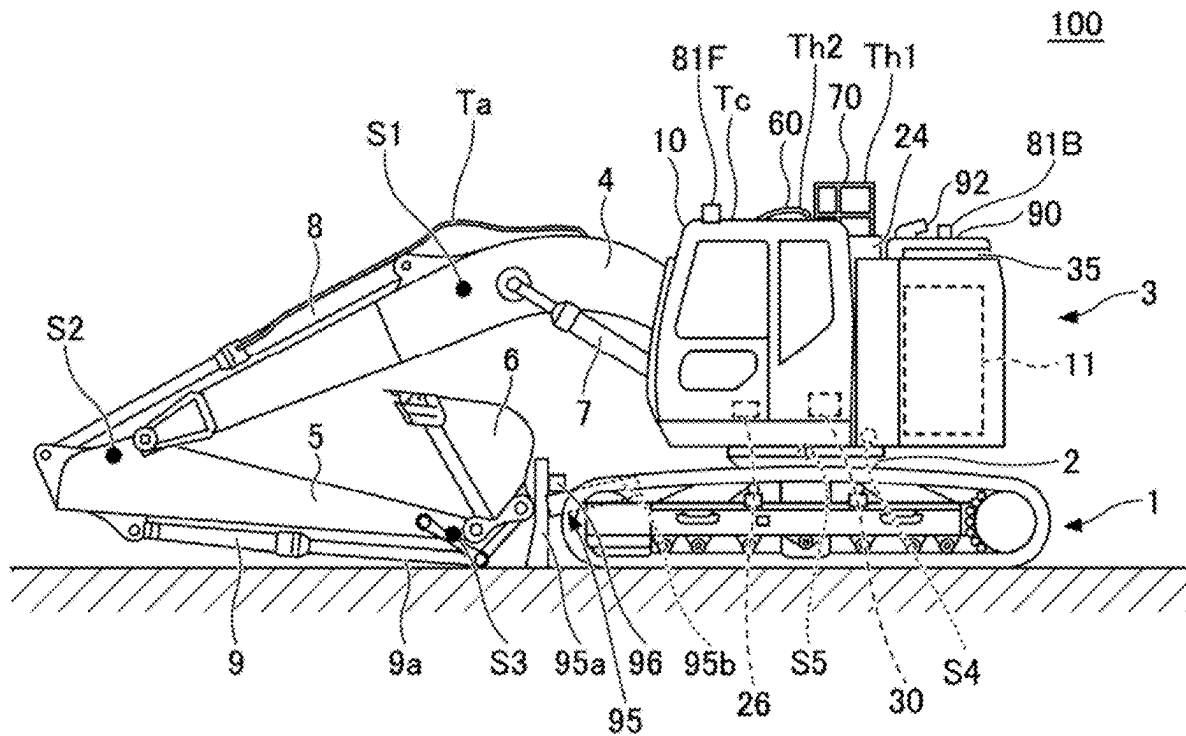


FIG. 1

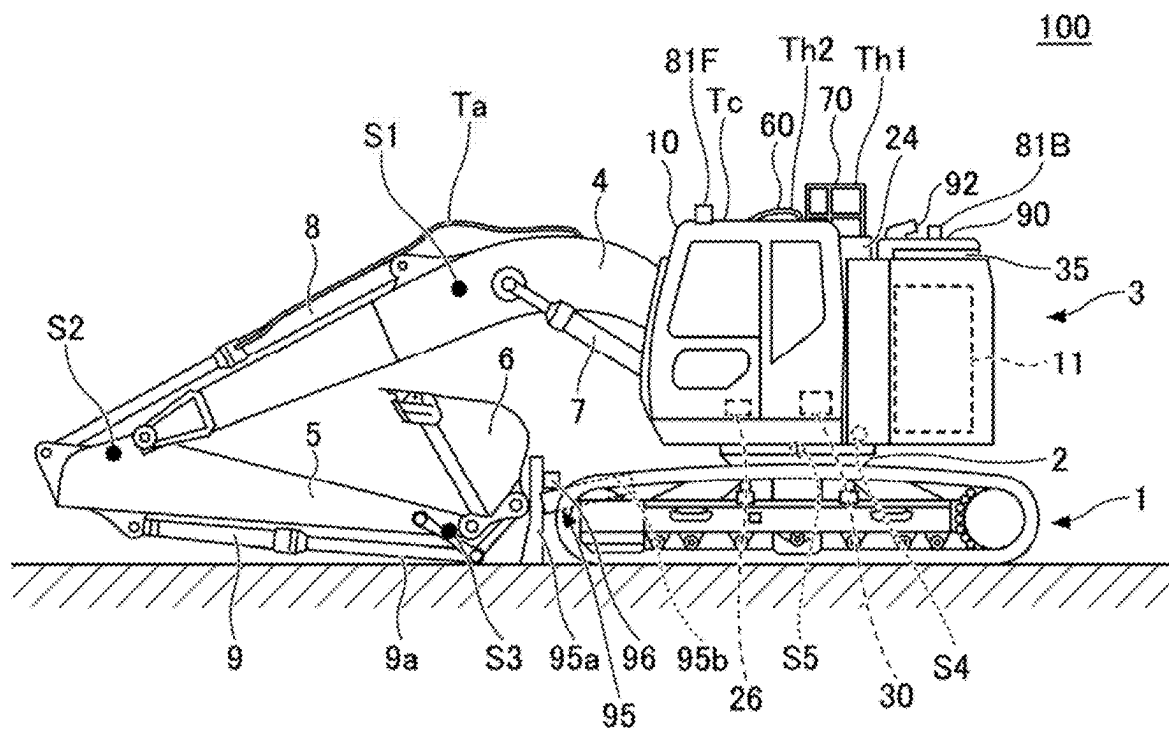


FIG.2

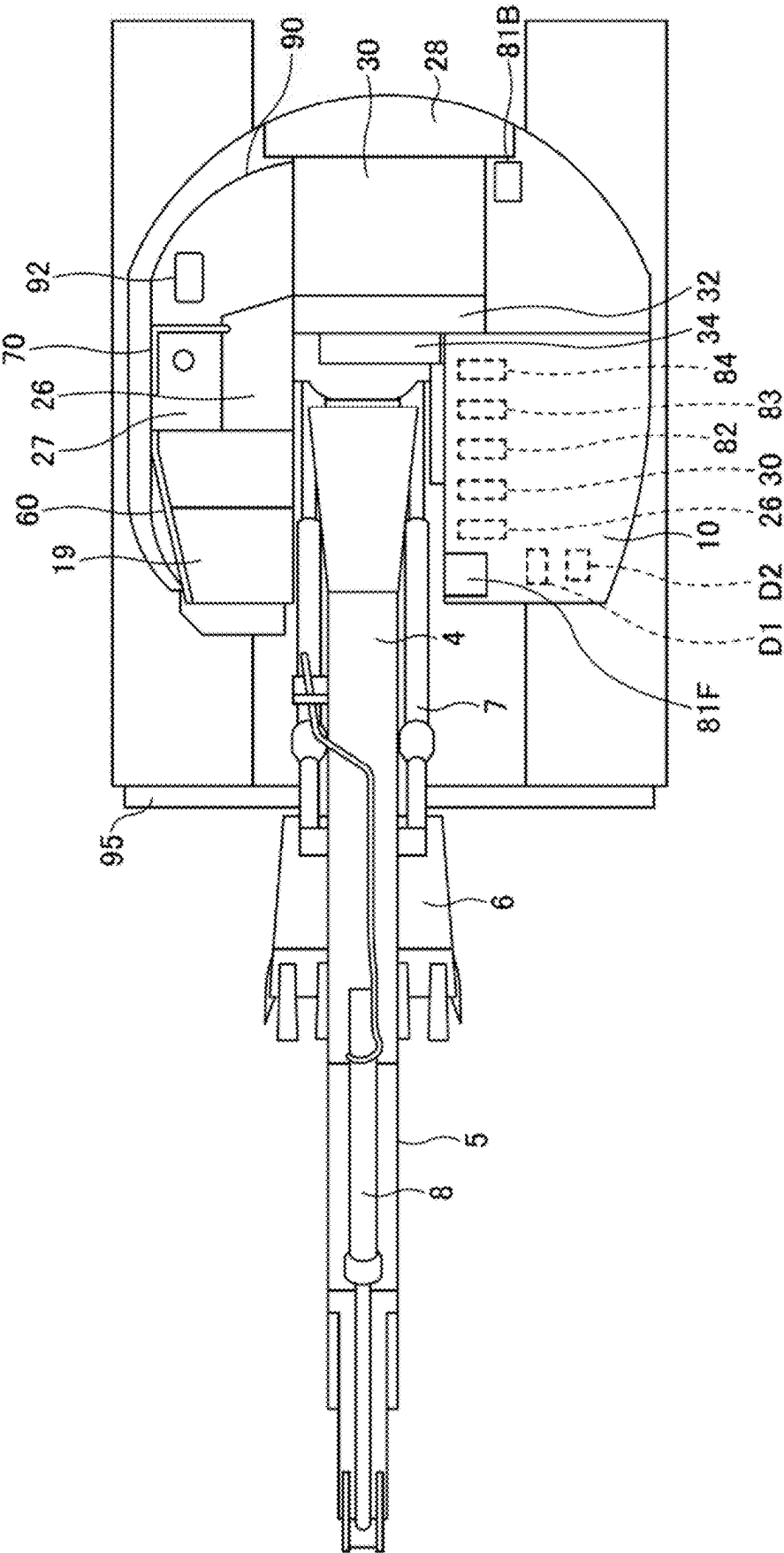


FIG.3

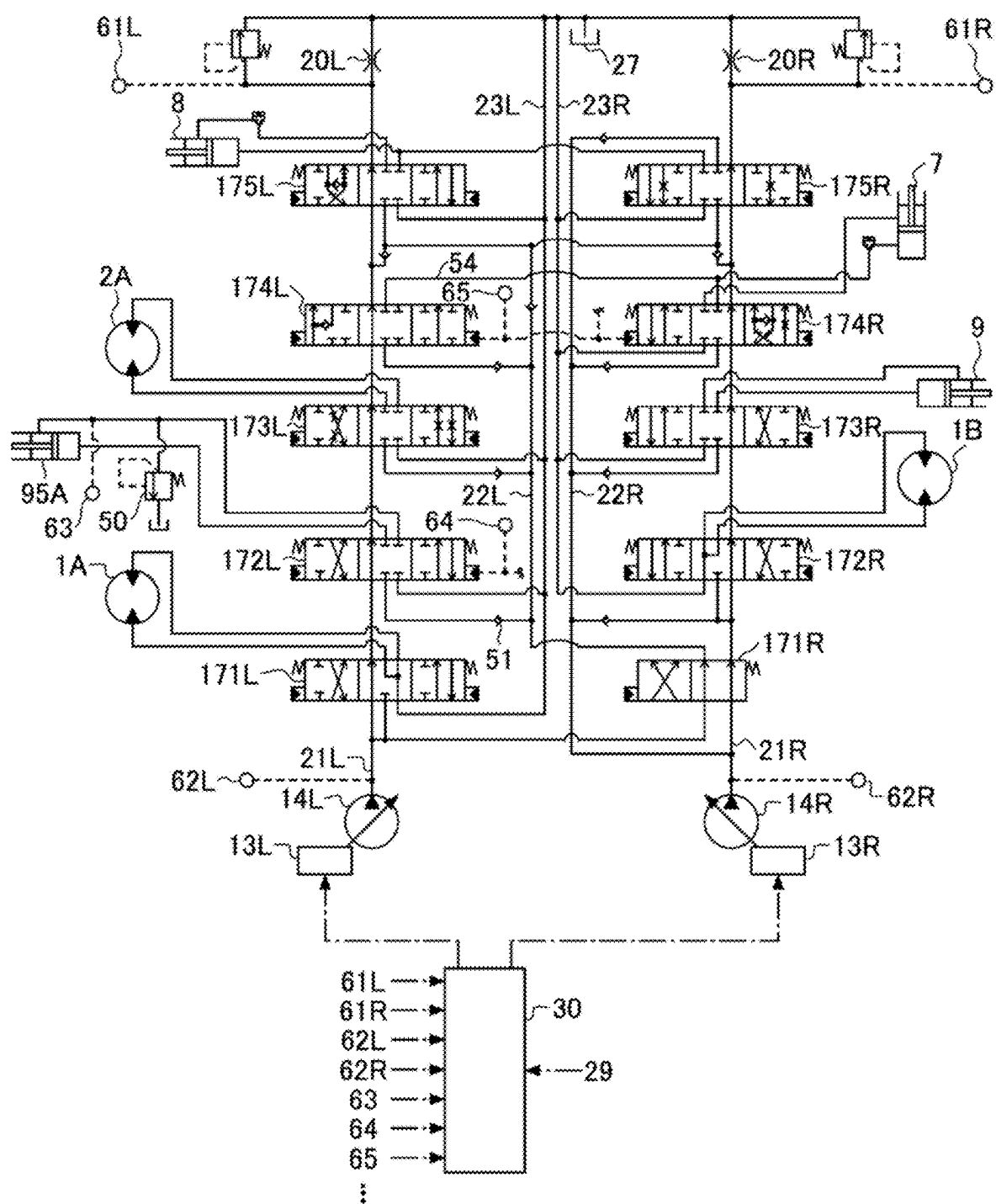


FIG.4A

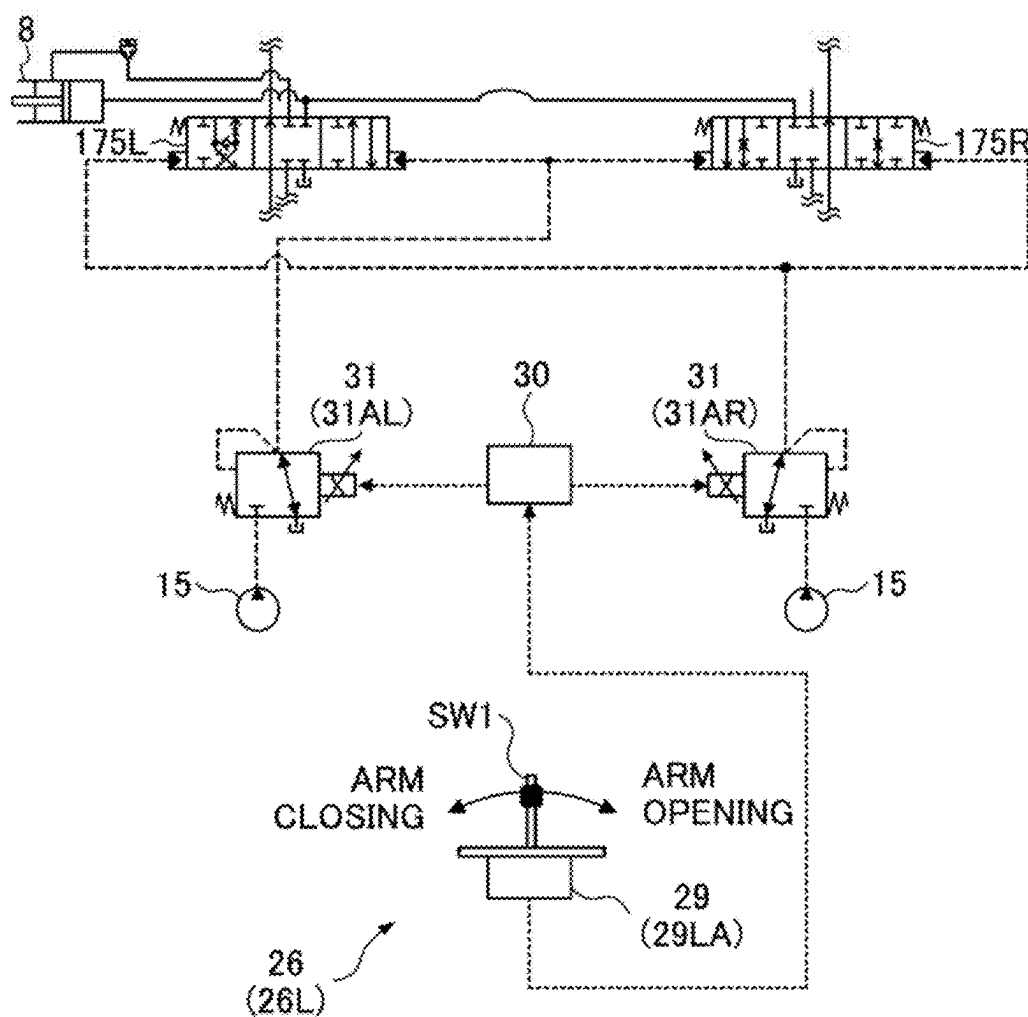


FIG.4B

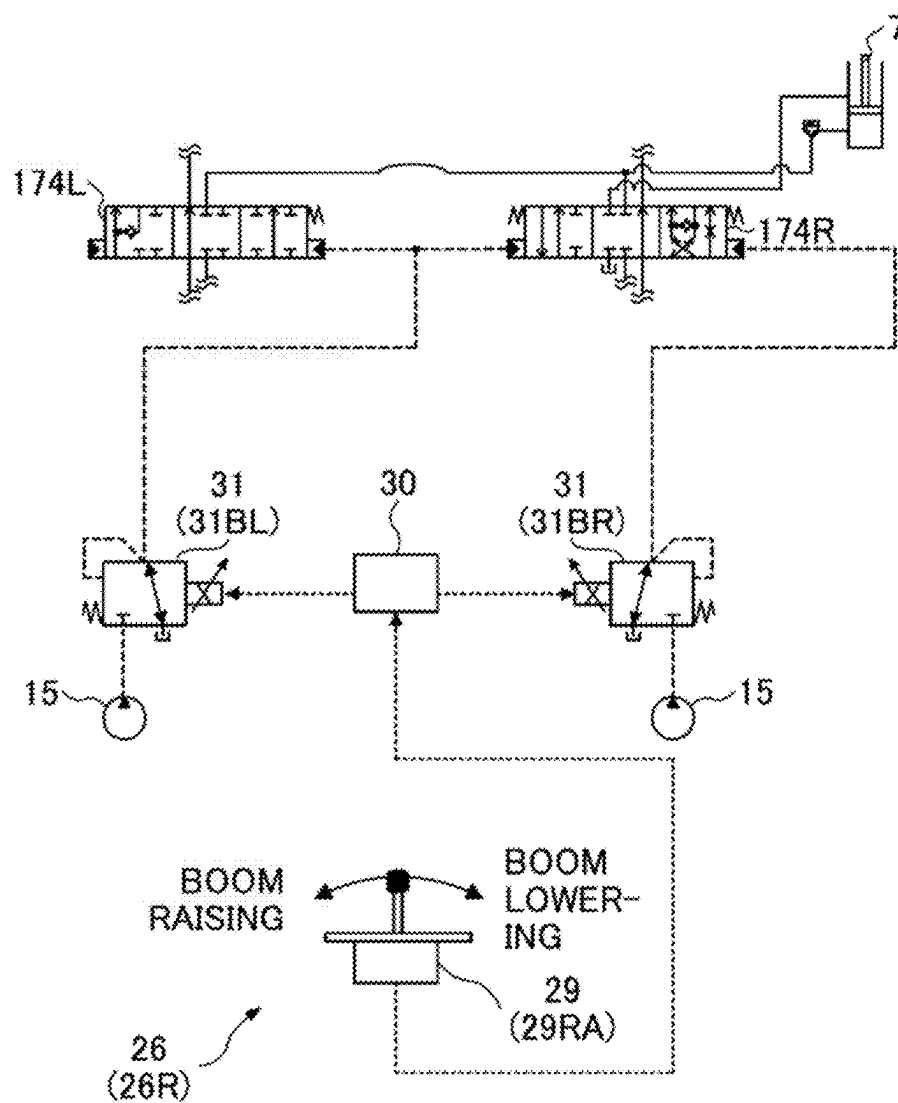


FIG.4C

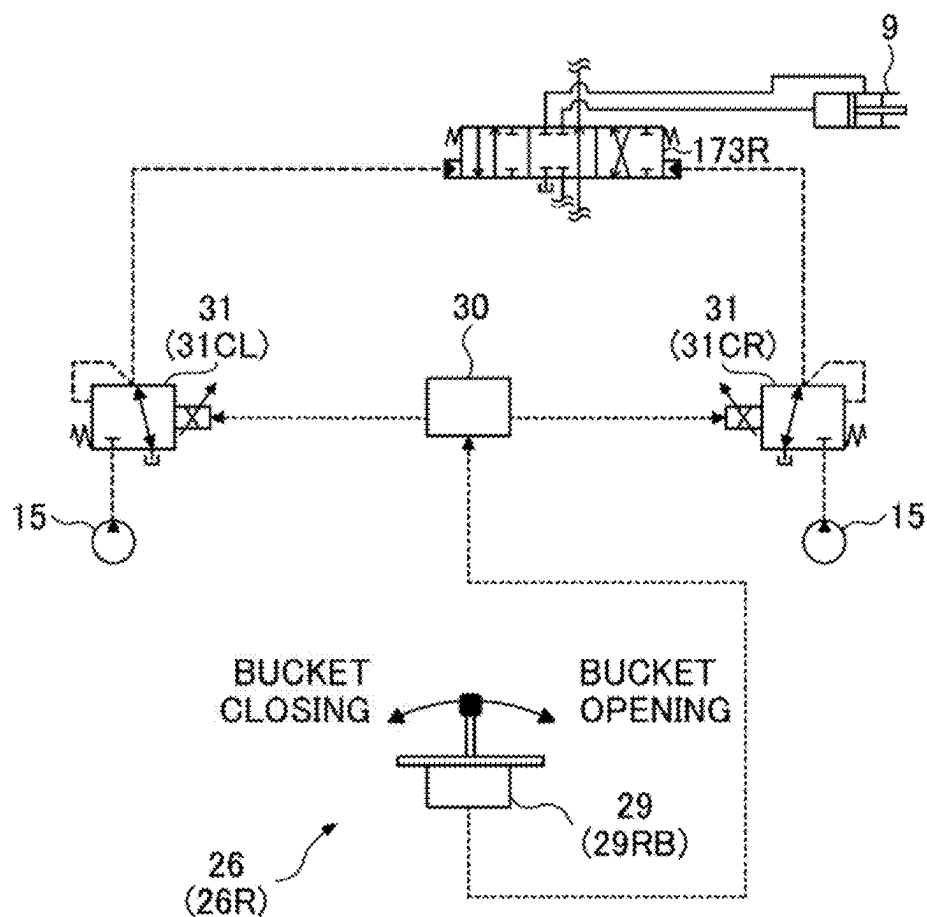
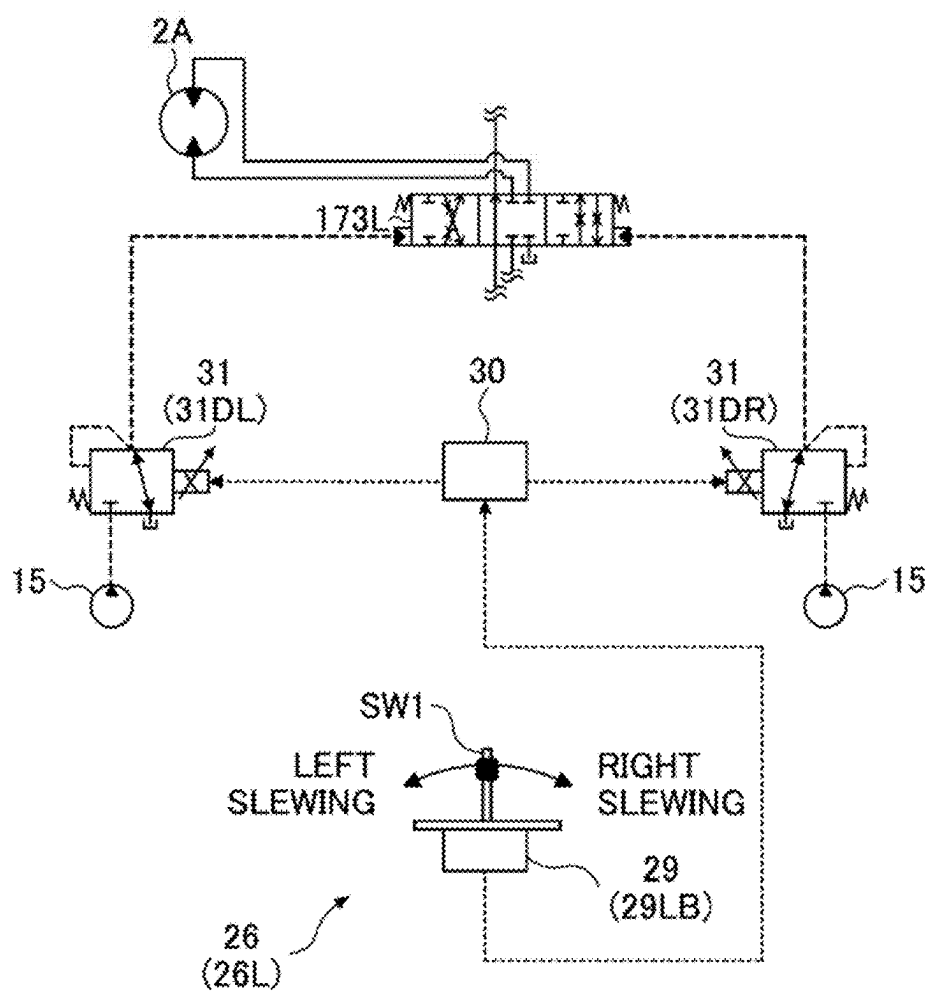


FIG.4D



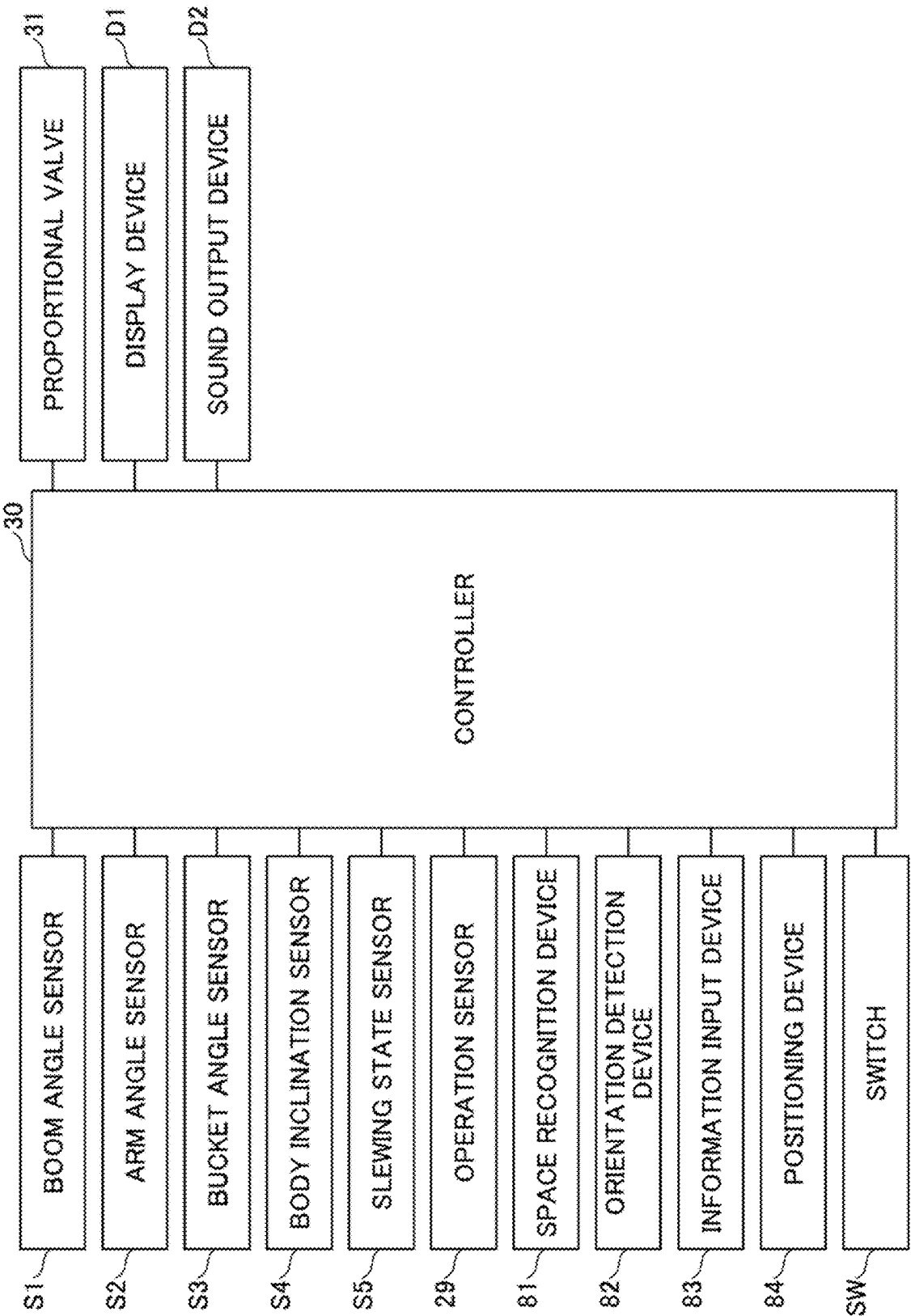


FIG.5

FIG. 6

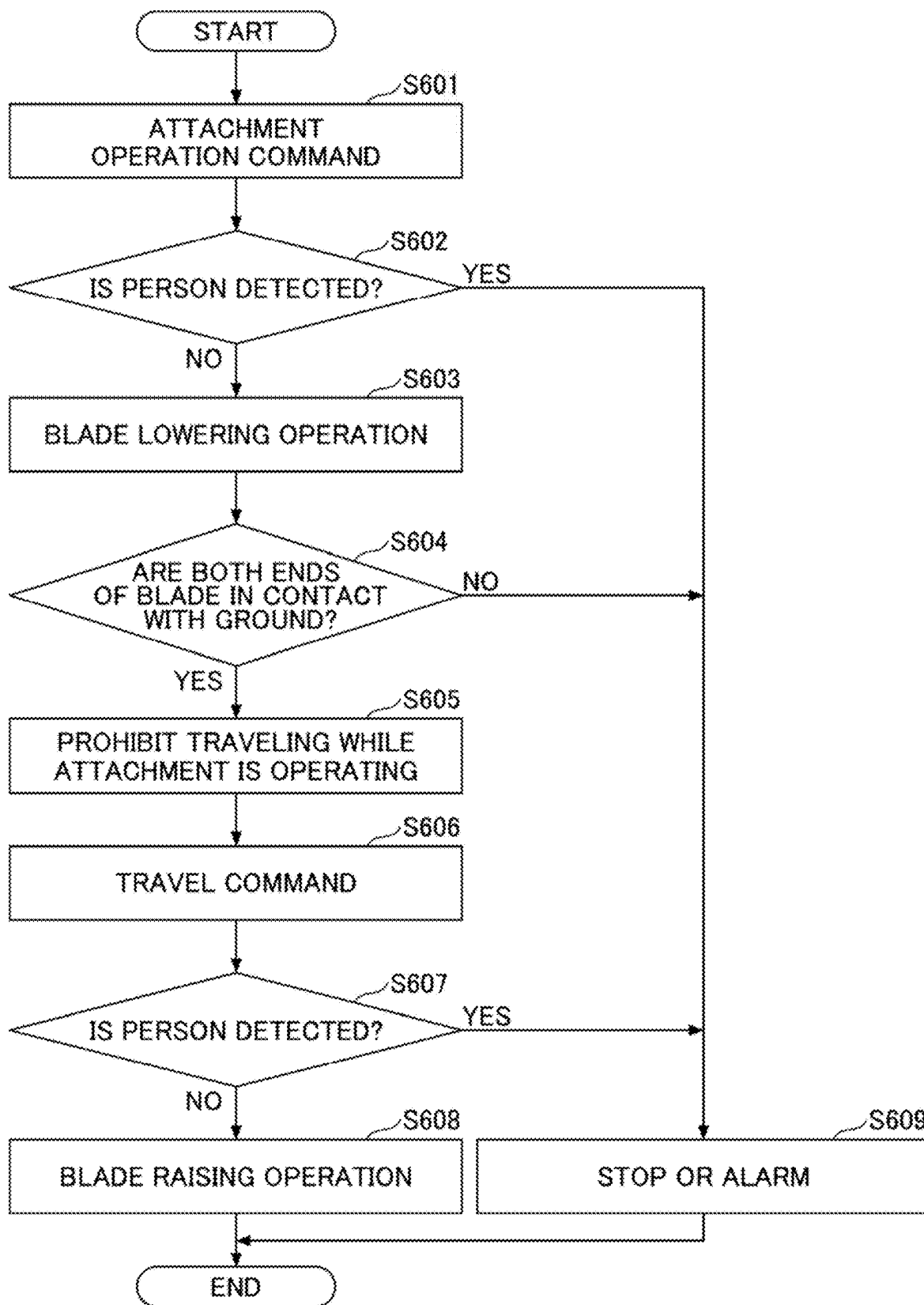


FIG. 7A

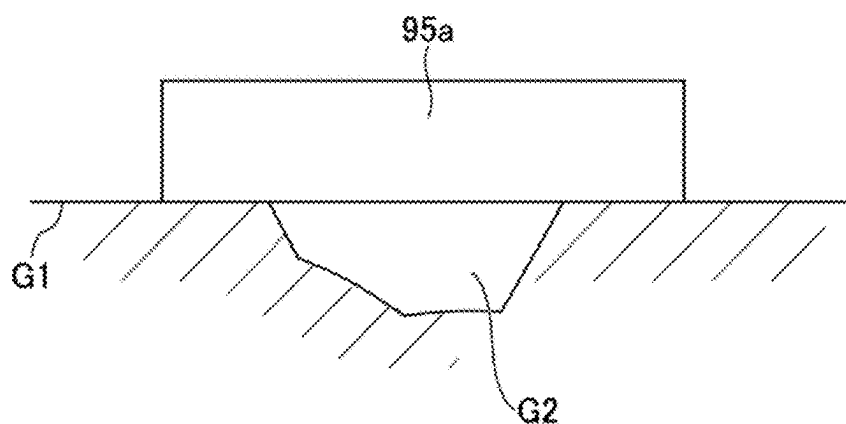
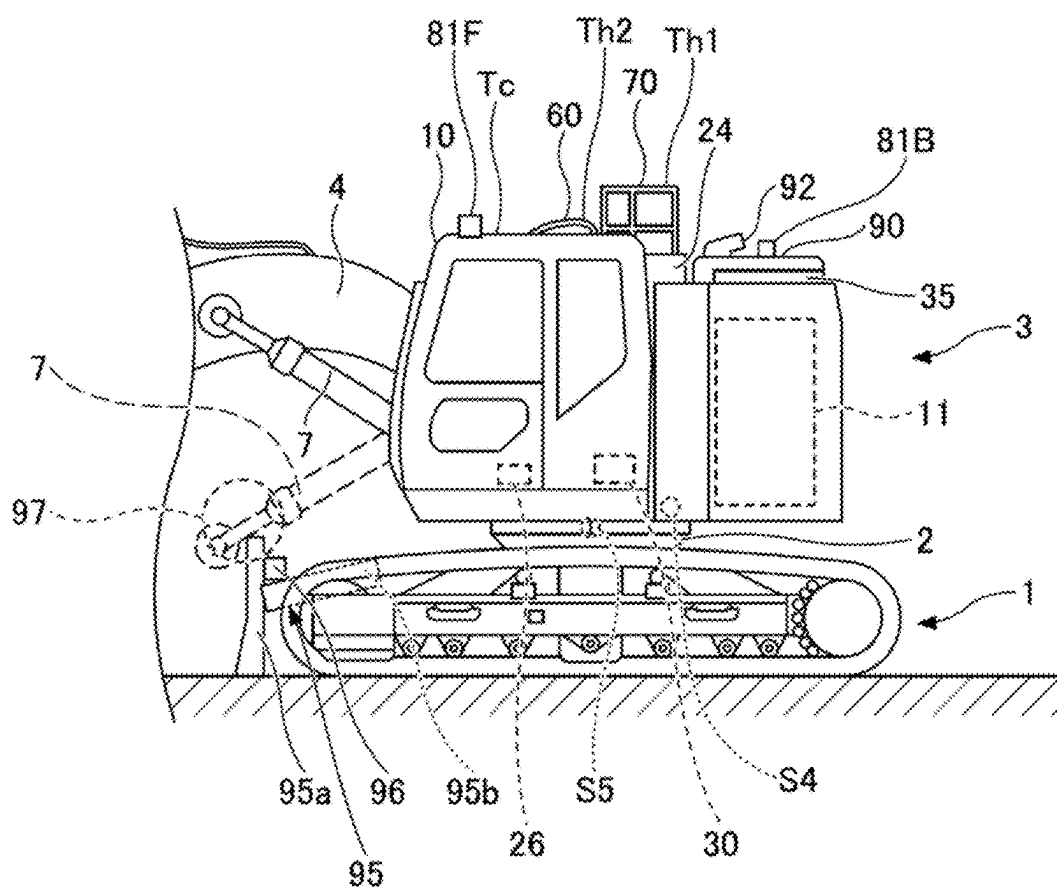


FIG. 7B



CONTROL DEVICE FOR EXCAVATOR

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation application filed under 35 U.S.C. 111(a) claiming benefit under 35 U.S.C. 120 and 365(c) of PCT International Application No. PCT/JP2023/040560, filed on Nov. 10, 2023, and designating the U.S., which claims priority to Japanese Patent Application No. 2022-187326 filed on Nov. 24, 2022. The entire contents of the foregoing applications are incorporated herein by reference.

BACKGROUND

Technical Field

[0002] The present disclosure relates to a control device for an excavator.

Description of Related Art

[0003] Conventionally, an excavator provided with a blade for performing a ground leveling operation is known. Further, in this excavator, the blade is in contact with the ground when the excavation work is performed, thereby preventing the excavator from falling.

SUMMARY

[0004] According to an embodiment of the present disclosure, a control device for an excavator is provided. The excavator includes an upper slewing body, an attachment provided on the upper slewing body, a slewing mechanism, a lower traveling body, and a blade provided on the lower traveling body. The control device includes:

[0005] circuitry configured to receive an input of an operation command for instructing an operation of the attachment, determine whether a person is detected around the excavator, and cause the excavator to perform an operation of lowering the blade when the person is not detected.

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. 1 is a side view illustrating a small slewing excavator.

[0007] FIG. 2 is a top view illustrating the small slewing excavator.

[0008] FIG. 3 is a diagram illustrating a configuration example of a hydraulic circuit mounted on the excavator.

[0009] FIG. 4A is a diagram illustrating an extracted portion of a hydraulic system relating to operation of an arm cylinder.

[0010] FIG. 4B is a diagram illustrating an extracted portion of the hydraulic system relating to operation of a boom cylinder.

[0011] FIG. 4C is a diagram illustrating an extracted portion of the hydraulic system relating to operation of a bucket cylinder.

[0012] FIG. 4D is a diagram illustrating an extracted portion of the hydraulic system relating to operation of a Slewing hydraulic motor.

[0013] FIG. 5 is a block diagram illustrating an example of a configuration related to a machine guidance function and a machine control function of the excavator.

[0014] FIG. 6 is a flowchart illustrating operation of the excavator.

[0015] FIG. 7A is a view illustrating a situation when a blade is lowered.

[0016] FIG. 7B is another view illustrating a situation when a blade is lowered.

EMBODIMENT OF THE INVENTION

[0017] In the above-described conventional excavator, an operation of lowering the blade is performed so as to bring the blade into contact with the ground every time the excavation work is performed. Further, in the case of the conventional excavator, when the excavator travels after performing the excavation work, an operation of raising the blade is required in order to protect the traveling surface. Therefore, in the conventional excavator, the operation of lowering and raising the blade is required every time the excavation work is performed, which is troublesome.

[0018] In view of the above-described point, it is an object of the present disclosure to improve operability.

[0019] According to an aspect of embodiments, operability can be improved.

[0020] Embodiments of the present disclosure will be described with reference to the accompanying drawings. FIG. 1 is a side view illustrating a small slewing excavator, and FIG. 2 is a top view illustrating the small slewing excavator.

[0021] Hereinafter, in the present specification, the small slewing excavator may be simply referred to as an “excavator”. An upper slewing body 3 is mounted on a lower traveling body 1 of the excavator 100 via a slewing mechanism 2. A boom 4 is attached to the upper slewing body 3. An arm 5 is attached to the distal end of the boom 4, and a bucket 6 as an end attachment is attached to the distal end of the arm 5. A slope bucket, a dredging bucket, or the like may be used as the end attachment.

[0022] The boom 4, the arm 5, and the bucket 6 constitute an attachment as an example of an attachment, and are hydraulically driven by a boom cylinder 7, an arm cylinder 8, and a bucket cylinder 9, respectively.

[0023] The upper slewing body 3 is provided with a cabin 10 and a power source such as an engine 11. In the cabin 10, an operator's seat, various operation devices 26 required for the excavator 100, and a controller 30 configured to control the driving of the excavator 100 are installed.

[0024] In FIG. 1, when a piston 9a of the bucket cylinder 9 mounted on the arm 5 is installed on a surface at the same height as the installation surface of the crawler, an uppermost portion Th1 of a handrail 70 mounted on a fuel tank 24 is at the highest position in the excavator 100. That is, the uppermost portion Th1 of the handrail 70 is higher than an uppermost portion Th2 of a handrail 60, an uppermost portion Tc of the cabin 10, and a pipe Ta to the arm cylinder 8 mounted on the boom 4. The upper surface of the fuel tank 24 is higher than the muffler cover 90.

[0025] When the handrail 70 is rotated as described later, the uppermost portion Th1 of the handrail 70 is lower than any of the uppermost portion Th2 of the handrail 60, the uppermost portion Tc of the cabin 10, and the pipe Ta to the arm cylinder 8 mounted on the boom 4.

[0026] In the excavator 100 according to the present embodiment, the lower traveling body 1 is provided with a blade 95 used for ground leveling work or the like. The blade 95 has a tip portion 95a and a support portion 95b, and the

blade is driven by a blade cylinder (not illustrated) provided in the lower traveling body 1. More specifically, the blade cylinder extends and contracts in response to the operation of the operator to move the blade 95 up and down. In other words, the excavator 100 performs a blade lowering operation of lowering the blade 95 and a blade raising operation of raising the blade 95 in response to the operation of the operator.

[0027] The excavator 100 also includes a blade angle sensor 96 for detecting the angle of the blade 95. The angle of the blade 95 may be, for example, an angle with respect to the horizontal plane of the support portion 95b.

[0028] The blade angle sensor 96 of the present embodiment transmits a detected signal to the controller 30 of the excavator 100 by wireless communication. The blade angle sensor 96 of the present embodiment may transmit a signal by electric power supplied by vibration power generation, solar power generation, or the like.

[0029] When the excavator 100 according to the present embodiment receives an input of an operation for instructing an operation of the attachment, the controller 30 performs a blade lowering operation for lowering the blade 95, and causes the blade 95 to contact the ground, thereby prohibiting the excavator 100 from traveling.

[0030] When the operation for the attachment is ended and the operation for instructing the traveling is received, the excavator 100 according to the present embodiment performs the blade raising operation for raising the blade 95 by the controller 30. In other words, the controller 30 performs the blade raising operation when no operation command is input to the attachment and a travel command for instructing the excavator to travel is input.

[0031] In the present embodiment, since the blade 95 is controlled by the controller 30 in this manner, the operator does not need to perform an operation of lowering the blade 95 every time the excavation work is performed, and the operation is thus simplified.

[0032] In the present embodiment, when the excavation work is performed, the blade 95 is lowered and is brought into contact with the ground, and thus it is possible to improve the stability of the body of the excavator 100 during the excavation work. Further, in the present embodiment, in a state where the blade 95 is in contact with the ground, the controller 30 prohibits the excavator 100 from traveling. Therefore, it is possible to prevent the excavator 100 from traveling with the blade 95 being lowered. The processing of the controller 30 will be described in detail later.

[0033] In the excavator 100 according to the present embodiment, as illustrated in FIG. 2, the boom 4 is rotatably supported near the center of the upper slewing body 3. The cabin 10 is installed on the front side of the upper slewing body 3 and on the left side of the boom 4. A urea water tank cover 19 for covering a urea water tank (described later) is attached to the front side of the upper slewing body 3 and the right side of the boom 4. The fuel tank 24 and a hydraulic fluid tank 27 are disposed behind the urea water tank cover 19. The fuel tank 24 is disposed on the outer side, and the hydraulic fluid tank 27 is disposed on the inner side of the fuel tank 24.

[0034] The upper surface of the urea water tank cover 19 and the upper surface of the hydraulic fluid tank 27 are used as a passage for a worker to pass during maintenance or the like. On the other hand, the upper surface of the fuel tank 24 is located at a position higher than the upper surface of the

hydraulic fluid tank 27 as described later, and is not used as a passage for the operator to pass.

[0035] A handrail 60 is attached to the outer peripheral portion of the upper slewing body 3 from the urea water tank cover 19 to the fuel tank 24. The handrail 60 is a rail for protecting the worker from falling when the worker climbs on the upper slewing body 3. The upper end of the handrail 60 extends to the end of the fuel tank 24. A handrail 70 extends from the position where the handrail 60 terminates in the direction in which the handrail 60 extends. The handrail 70 is attached to the upper surface of the fuel tank 24 and is bent inward along the corner of the upper surface of the fuel tank 24. The handrail 70 is provided at a position where the operator can hold the handrail 70 when the operator climbs the upper surface of the hydraulic fluid tank 27.

[0036] A muffler cover 90 for covering an exhaust gas treatment device described later is provided behind the fuel tank 24 and the hydraulic fluid tank 27. An exhaust pipe (muffler) 92 extending from the exhaust gas treatment device protrudes from a portion of the muffler cover 90 near the fuel tank 24.

[0037] The urea water tank cover 19, the fuel tank 24, the hydraulic fluid tank 27, the handrail 60, the handrail 70, and the muffler cover 90 are disposed on the right side of the upper slewing body 3.

[0038] As described above, the boom 4 is rotatably attached to the center of the upper slewing body 3. The shape of the rear portion of the upper slewing body 3 is defined by the slewing radius of the excavator, and is an arc shape having the slewing radius. A counterweight 28 is disposed at the center of the rear portion of the upper slewing body 3. The engine 11 is disposed and fixed in a space between the counterweight 28 and the boom 4. In FIGS. 1 and 2, the engine 11 is not illustrated because an engine hood 35 is provided to cover the upper portion of the engine 11.

[0039] A passage 32 is provided on the front side of the engine hood 35, and an operator passes the passage 32 when the engine hood 35 is opened to perform maintenance work of the engine 11. A working scaffold 34 is provided at a position lower than the passage 32 on the front side of the passage 32. The working scaffold 34 is provided so as to cover the slewing motor disposed beside the hydraulic fluid tank 27, and also functions as a slewing motor cover. When the worker climbs the upper passage 32, the worker first climbs the lower working scaffold 34 and then climbs the higher passage 32. At this time, the worker can easily climb from the working scaffold 34 to the passage 32 by placing his/her hand on a portion of the end portion of the handrail 70 extending in the vertical direction.

[0040] The excavator 100 according to the present embodiment further includes a boom angle sensor S1, an arm angle sensor S2, a bucket angle sensor S3, a body inclination sensor S4, and a slewing state sensor S5. The excavator 100 according to the present embodiment includes a space recognition device 81, an orientation detection device 82, an input device 83, a positioning device 84, a display device D1, and a sound output device D2.

[0041] The boom angle sensor S1 is attached to the boom 4 and detects an elevation angle of the boom 4 with respect to the upper slewing body 3 (hereinafter, referred to as a "boom angle"), for example, an angle formed by a straight

line connecting fulcrums at both ends of the boom 4 with respect to a slewing plane of the upper slewing body 3 in a side view.

[0042] The boom angle sensor S1 may include, for example, a rotary encoder, an accelerometer, a gyro sensor (angular velocity sensor), a six axis sensor, an inertial measurement unit (IMU), and the like. Hereinafter, the same applies to the arm angle sensor S2, the bucket angle sensor S3, and the body inclination sensor S4. A detection signal corresponding to the boom angle by the boom angle sensor S1 is incorporated into the controller 30.

[0043] The arm angle sensor S2 is attached to the arm 5 and detects a rotation angle of the arm 5 with respect to the boom 4 (hereinafter, referred to as an “arm angle”), for example, an angle formed by a straight line connecting fulcrums at both ends of the arm 5 with respect to a straight line connecting fulcrums at both ends of the boom 4 in a side view. A detection signal corresponding to the arm angle by the arm angle sensor S2 is incorporated into the controller 30.

[0044] The bucket angle sensor S3 is attached to the bucket 6 and detects a rotation angle of the bucket 6 with respect to the arm 5 (hereinafter, referred to as a “bucket angle”), for example, an angle formed by a straight line connecting a fulcrum and a tip (a blade edge) of the bucket 6 with respect to a straight line connecting fulcrums at both ends of the arm 5 in a side view. A detection signal corresponding to the bucket angle by the bucket angle sensor S3 is incorporated into the controller 30.

[0045] In the present embodiment, the above-described various angle sensors may include an operation amount detection part that detects an operation amount, and the various angle sensors may calculate an angle based on the detected operation amount.

[0046] The body inclination sensor S4 detects an inclination state of the body (e.g., the upper slewing body 3) with respect to the horizontal plane. The body inclination sensor S4 is attached to, for example, the upper slewing body 3, and detects inclination angles (hereinafter, referred to as a “front-rear inclination angle” and a “left-right inclination angle”) around two axes in the front-rear direction and the left-right direction of the excavator 100 (i.e., the upper slewing body 3). The body inclination sensor S4 may include, for example, an accelerometer, a gyroscope (angular velocity sensor), a six axis sensor, an IMU, and the like. The controller 30 receives a detection signal corresponding to the inclination angle (the front-rear inclination angle and the left-right inclination angle) from the body inclination sensor S4.

[0047] The slewing state sensor S5 is attached to the upper slewing body 3 and outputs detection information related to the slewing state of the upper slewing body 3. The slewing state sensor S5 detects, for example, a slewing angular velocity and a slewing angle of the upper slewing body 3. The slewing state sensor S5 includes, for example, a gyro sensor, a resolver, a rotary encoder, and the like. Note that, when the body inclination sensor S4 includes a gyro sensor, a six axis sensor, an IMU, or the like capable of detecting angular velocities about three axes, the slewing state (e.g., slewing angular velocities) of the upper slewing body 3 may be detected based on the detection signal of the body inclination sensor S4. In this case, the slewing state sensor S5 may be omitted.

[0048] The space recognition device 81 is configured to recognize an object existing in a three dimensional space around the excavator 100 and measure (calculate) a positional relationship such as a distance from the space recognition device 81 or the excavator 100 to the recognized object. The space recognition device 81 may include, for example, an ultrasonic sensor, a millimeter wave radar, a monocular camera, a stereo camera, a light detecting and ranging (LIDAR), a range image sensor, and an infrared sensor.

[0049] In the present embodiment, the space recognition device 81 includes a front recognition sensor 81F attached to the front end of the upper surface of the cabin 10 and a rear recognition sensor 81B attached to the rear end of the upper surface of the upper slewing body 3. The space recognition device 81 of the present embodiment may include a left recognition sensor attached to the left end of the upper surface of the upper slewing body 3 and a right recognition sensor attached to the right end of the upper surface of the upper slewing body 3. Further, an upper recognition sensor that recognizes an object existing in the space above the upper slewing body 3 may be attached to the excavator 100.

[0050] The orientation detection device 82 detects information (e.g., a slewing angle of the upper slewing body 3 with respect to the lower traveling body 1) related to a relative relationship between the orientation of the upper slewing body 3 and the orientation of the lower traveling body 1.

[0051] The orientation detection device 82 may include, for example, a combination of a geomagnetic sensor attached to the lower traveling body 1 and a geomagnetic sensor attached to the upper slewing body 3. The orientation detection device 82 may include a combination of a GNSS receiver attached to the lower traveling body 1 and a GNSS receiver attached to the upper slewing body 3. The orientation detection device 82 may include a rotary encoder, a rotary position sensor, or the like capable of detecting the relative slewing angle of the upper slewing body 3 with respect to the lower traveling body 1, that is, the slewing state sensor S5 described above, and may be attached to, for example, a center joint provided in association with the slewing mechanism 2 that implements the relative rotation between the lower traveling body 1 and the upper slewing body 3.

[0052] The orientation detection device 82 may include a camera attached to the upper slewing body 3. In this case, the orientation detection device 82 performs known image processing on an image (input image) captured by the camera attached to the upper slewing body 3, thereby detecting an image of the lower traveling body 1 included in the input image.

[0053] In the case of a configuration in which the upper slewing body 3 is slewably driven by a motor instead of the slewing hydraulic motor 2A, the orientation detection device 82 may be a resolver.

[0054] The input device 83 is provided within reach of an operator seated in the cabin 10, receives various operation inputs from the operator, and outputs signals corresponding to the operation inputs to the controller 30. For example, the input device 83 may include a touch panel mounted on a display of the display device D1 that displays various information images.

[0055] Further, for example, the input device 83 can include a button switch, a lever, a toggle, or the like installed

around the display device **D1**. The input device **83** may include a knob switch provided on the operation device **26** (e.g., a switch **SW** provided on a left operation lever **26L**). A signal corresponding to the operation content on the input device **83** is incorporated into the controller **30**.

[0056] The switch **SW** is, for example, a push button switch provided at the tip of the left operation lever **26L**. The operator can operate the left operation lever **26L** while pressing the switch **SW**. The switch **SW** may be provided on a right operation lever **26R** or at another position in the cabin **10**.

[0057] A positioning device **84** measures the position and orientation of the upper slewing body **3**. The positioning device **84** is, for example, a global navigation satellite system (GNSS) compass, and detects the position and orientation of the upper slewing body **3**. Detection signals corresponding to the position and orientation of the upper slewing body **3** are incorporated into the controller **30**. Further, the function of detecting the orientation of the upper slewing body **3** among the functions of the positioning device **84** may be replaced by an orientation sensor attached to the upper slewing body **3**.

[0058] The display device **D1** is provided at a position easily visible from the operator seated in the cabin **10**, and displays various information images under the control of the controller **30**. The display device **D1** may be connected to the controller **30** via an in-vehicle communication network such as a controller area network (CAN) or may be connected to the controller **30** via a one-to-one dedicated line.

[0059] The sound output device **D2** is provided in the cabin **10**, for example, is connected to the controller **30**, and outputs sound under the control of the controller **30**. The sound output device **D2** is, for example, a speaker, a buzzer, or the like. The sound output device **D2** outputs various kinds of information in response to an audio command from the controller **30**.

[0060] Next, a configuration example of a hydraulic circuit mounted on the excavator **100** of FIG. 1 will be described with reference to FIG. 3. FIG. 3 is a diagram illustrating a configuration example of a hydraulic circuit mounted on the excavator. In FIG. 3, the high-pressure hydraulic line, the pilot line, and the electric control system are indicated by a solid line, a broken line, and a dash dot line, respectively.

[0061] The main pumps **14L** and **14R** are variable displacement hydraulic pumps driven by the engine **11**. In the present embodiment, the main pump **14L** circulates the hydraulic fluid to the hydraulic fluid tank **27** through a center bypass oil passage **21L** passing through each of the control valves **171L** to **175L** constituting the control valve **17**. The main pump **14L** can supply the hydraulic fluid to each of the control valves **172L** to **175L** through a parallel oil passage **22L** extending in parallel to the center bypass oil passage **21L**.

[0062] Similarly, the main pump **14R** circulates the hydraulic fluid to the hydraulic fluid tank **27** through a center bypass oil passage **21R** passing through each of the control valves **171R** to **175R** constituting the control valve **17**. The main pump **14R** can supply the hydraulic fluid to each of the control valves **172R** to **175R** through a parallel oil passage **22R** extending in parallel to the center bypass oil passage **21R**. Hereinafter, the main pump **14L** and the main pump **14R** may be collectively referred to as a "main pump **14**".

The same applies to the other components configured as a pair of left and right components.

[0063] The control valve **171L** is a spool valve that switches the flow of the hydraulic fluid discharged from the main pump **14L** to supply the hydraulic fluid to the left traveling hydraulic motor **1A** when a left travel lever (not illustrated) is operated.

[0064] The control valve **171R** is a spool as a straight traveling control valve. In the present embodiment, the control valve **171R** has a first position and a second position. Specifically, the first position includes a flow passage that communicates the main pump **14L** with the parallel oil passage **22L** and a flow passage that communicates the main pump **14R** with the control valve **172R**. The second position includes a flow passage that communicates the main pump **14R** with the parallel oil passage **22L** and a flow passage that communicates the main pump **14L** with the control valve **172R**.

[0065] The control valve **172L** is a spool valve that switches the flow of the hydraulic fluid to supply the hydraulic fluid discharged from the main pump **14L** to a blade cylinder **95A** when the blade **95** is operated by the operation device **26**. In other words, the control valve **172L** is a spare control valve.

[0066] The control valve **172R** is a spool valve that switches the flow of the hydraulic fluid to supply the hydraulic fluid discharged by the main pump **14** to the right traveling hydraulic motor **1B** when the operation device **26** is operated. The operation device **26** will be described in detail later.

[0067] The control valve **173L** is a spool valve that switches the flow of the hydraulic fluid to supply the hydraulic fluid discharged by the main pump **14** to the slewing hydraulic motor **2A** when the slewing mechanism **2** is operated by the operation device **26**.

[0068] The control valve **173R** is a spool valve that switches the flow of the hydraulic fluid to supply the hydraulic fluid discharged by the main pump **14R** to the bucket cylinder **9** when the bucket **6** is operated by the operation device **26**.

[0069] The control valves **174L** and **174R** are spool valves that switch the flow of the hydraulic fluid to supply the hydraulic fluid discharged by the main pump **14** to the boom cylinder **7** when the boom **4** is operated by the operation device **26**. The control valve **174L** additionally supplies the hydraulic fluid to the boom cylinder **7** when the operation lever for operating the boom **4** is operated in the boom raising direction by a predetermined lever operation amount or more. The operation lever is part of the operation device **26**.

[0070] The control valves **175L** and **175R** are spool valves that switch the flow of the hydraulic fluid to supply the hydraulic fluid discharged by the main pump **14** to the arm cylinder **8** when the arm **5** is operated by the operation device **26**. The control valve **175R** additionally supplies the hydraulic fluid to the arm cylinder **8** when the operation lever for operating the arm **5** is operated by a predetermined lever operation amount or more.

[0071] The hydraulic fluid flowing out from each of the left traveling hydraulic motor **1A**, the blade cylinder **95A**, the slewing hydraulic motor **2A**, and the arm cylinder **8** is discharged to the hydraulic fluid tank **27** through the return oil passage **23L**. Similarly, the hydraulic fluid flowing out from each of the right traveling hydraulic motor **1B**, the

bucket cylinder 9, and the boom cylinder 7 is discharged to the hydraulic fluid tank 27 through the return oil passage 23R. Further, a part of the hydraulic fluid flowing out from the arm cylinder 8 may be discharged to the hydraulic fluid tank 27 through a return oil passage 23R.

[0072] The center bypass oil passages 21L and 21R are provided with negative control throttles 20L and 20R between the control valves 175L and 175R on the most downstream side and the hydraulic fluid tank 27, respectively. Hereinafter, the negative control is abbreviated as “negative control”. The negative control throttles 20L and 20R restrict the flow of the hydraulic fluid discharged from the main pumps 14L and 14R to generate a negative control pressure upstream of the negative control throttles 20L and 20R.

[0073] The controller 30 performs the negative control by using the negative control pressure. Specifically, the lower the negative control pressures generated in the negative control throttles 20L and 20R, the more the discharge amounts of the main pumps 14L and 14R is increased. When the negative control pressures generated by the negative control throttles 20L and 20R exceed predetermined pressures, the discharge amounts of the main pumps 14L and 14R are reduced to predetermined lower limit values.

[0074] The relief valve 50 is a valve for controlling the pressure in the rod side oil chamber of the blade cylinder 95A to be equal to or lower than predetermined closing relief pressures.

[0075] The load check valve 51 is a valve for preventing the hydraulic fluid in the blade cylinder 95A from flowing back to the parallel oil passage 22L.

[0076] The negative control pressures generated upstream of the negative control throttles 20L and 20R are detected by the pressure sensors 61L and 61R, and the detected values are output to the controller 30 as electrical negative control pressure signals.

[0077] The pressure sensors 62L and 62R detect the discharge pressures of the main pumps 14L and 14R, and output the detected values to the controller 30 as electric discharge pressure signals.

[0078] The pressure sensor 63 detects the oil pressure in the rod side oil chamber of the blade cylinder 95A and outputs the detected value to the controller 30 as an electric blade rod pressure signal.

[0079] The pressure sensor 64 is one of the pressure sensors, which detects a pilot hydraulic pressure applied to the right pilot port of the control valve 172L, and outputs the detected value to the controller 30 as an electric signal.

[0080] The pressure sensor 65 is one of the pressure sensors, which detects the pilot pressure applied to the left side (boom raising side) pilot port of the control valve 174L and the right side (boom raising side) pilot port of control valve 174R (hereinafter referred to as “boom raising pilot pressure”), and outputs the detected value as an electrical boom raising pilot pressure signal to the controller 30.

[0081] The controller 30 receives outputs from the pressure sensors 61L, 61R, 62L, 62R, 63, 64, 65, and the like, and causes the CPU to execute a program for adjusting the discharge amounts of the main pumps 14L and 14R.

[0082] Further, when both the hydraulic actuators (e.g., the blade cylinder 95A) related to the main pump 14L and the hydraulic actuators (e.g., the boom cylinder 7) related to the main pump 14R are continuously operated by full lever/full pedal (e.g., an operation amount of 80% or more

when a neutral state of the lever/pedal is 0% and a maximum operation state is 100%), the controller 30 sets the discharge amount L1 of the main pump 14L and the discharge amount L2 of the main pump 14R to be the same. Hereinafter, this method is referred to as a “discharge amount synchronization method”.

[0083] Next, a configuration for the controller 30 to operate the actuators by the machine control function will be described with reference to FIGS. 4A to 4D. FIGS. 4A to 4D are diagrams in which a part of the hydraulic system is extracted. Specifically, FIG. 4A is a diagram illustrating an extracted portion of a hydraulic system relating to the operation of the arm cylinder 8, and FIG. 4B is a diagram illustrating an extracted portion of the hydraulic system relating to the operation of the boom cylinder 7. FIG. 4C is a diagram illustrating an extracted portion of the hydraulic system related to the operation of the bucket cylinder 9 is extracted, and FIG. 4D is a diagram illustrating an extracted portion of the hydraulic system related to the operation of the slewing hydraulic motor 2A is extracted.

[0084] As illustrated in FIGS. 4A to 4D, the hydraulic system includes proportional valves 31. The proportional valves 31 include proportional valves 31AL to 31DL and proportional valves 31AR to 31DR.

[0085] The proportional valves 31 each functions as a control valve for machine control. The proportional valves 31 are disposed in a conduit connecting pilot pump 15 and pilot ports of corresponding control valves in a control valve 17, and is configured to be able to change a flow passage area of the conduit.

[0086] In the present embodiment, the proportional valves 31 operate in response to a control command output from the controller 30. Therefore, the controller 30 can supply the hydraulic fluid discharged by the pilot pump 15 to the pilot ports of the corresponding control valves in the control valve 17 via the proportional valves 31, independent of the operation of the operation device 26 by the operator. The controller 30 can apply the pilot pressures generated by the proportional valves 31 to the pilot ports of the corresponding control valves.

[0087] With this configuration, even when an operation is not performed on a specific operation device 26, the controller 30 can operate a hydraulic actuator corresponding to the specific operation device 26. Further, even when an operation is being performed on the specific operation device 26, the controller 30 can forcibly stop the operation of the hydraulic actuator corresponding to the specific operation device 26.

[0088] For example, as illustrated in FIG. 4A, the left operation lever 26L is used to operate the arm 5. Specifically, the left operation lever 26L uses the hydraulic fluid discharged from the pilot pump 15 to apply pilot pressures corresponding to the operation in the front-rear direction to the pilot ports of the control valve 175L and the control valve 175R. Specifically, when the left operation lever 26L is operated in the arm closing direction (rearward direction), the left operation lever 26L applies a pilot pressure corresponding to the operation amount to the right pilot port of the control valve 175L and the left pilot port of the control valve 175R. When the left operation lever 26L is operated in the arm opening direction (forward direction), the left operation lever 26L applies a pilot pressure corresponding to the operation amount to the left pilot port of the control valve 175L and the right pilot port of the control valve 175R.

[0089] The operation device 26 is provided with a switch SW. In the present embodiment, the switch SW includes a switch SW1 and another switch provided at the tip of a travel lever (not illustrated).

[0090] The switch SW1 is a push button switch provided at the tip of the left operation lever 26L. The operator can operate the left operation lever 26L while pressing the switch SW1. The switch SW1 may be provided on the right operation lever 26R or may be provided at another position in the cabin 10.

[0091] The other switch is a push button switch provided at the tip of the left traveling lever. The operator can operate the left travel lever while pressing another switch. The other switch may be provided on the right travel lever included in the operation device 26, or may be provided at another position in the cabin 10.

[0092] The operation sensors 29 include operation sensors 29LA and 29LB, and operation sensors 29RA and 29RB. The operation sensor 29LA detects the operation content of the left operation lever 26L in the front-rear direction by the operator, and outputs the detected value to the controller 30.

[0093] The proportional valve 31AL operates in response to a control command (electric current command) output from the controller 30. The pilot pressure is adjusted by the hydraulic fluid introduced from the pilot pump 15 to the right pilot port of the control valve 175L and the left pilot port of the control valve 175R via the proportional valve 31AL.

[0094] The proportional valve 31AR operates in response to a control command (electric current command) output from the controller 30. The pilot pressure is adjusted by the hydraulic fluid introduced from the pilot pump 15 to the left pilot port of the control valve 175L and the right pilot port of the control valve 175R via the proportional valve 31AR. The proportional valve 31AL can adjust the pilot pressure so that the control valve 175L and the control valve 175R can be stopped at any positions. Similarly, the proportional valve 31AR can adjust the pilot pressure so that the control valve 175L and the control valve 175R can be stopped at any positions.

[0095] With this configuration, the controller 30 can supply the hydraulic fluid discharged by the pilot pump 15 to the right pilot port of the control valve 175L and the left pilot port of the control valve 175R via the proportional valve 31AL in response to the arm closing operation by the operator. Further, the controller 30 can supply the hydraulic fluid discharged from the pilot pump 15 to the right pilot port of the control valve 175L and the left pilot port of the control valve 175R via the proportional valve 31AL, independently of the arm closing operation by the operator. That is, the controller 30 can close the arm 5 in response to the arm closing operation by the operator or independently of the arm closing operation by the operator.

[0096] Further, the controller 30 can supply the hydraulic fluid discharged from the pilot pump 15 to the left pilot port of the control valve 175L and the right pilot port of the control valve 175R via the proportional valve 31AR in response to the arm opening operation by the operator. Further, the controller 30 can supply the hydraulic fluid discharged from the pilot pump 15 to the left pilot port of the control valve 175L and the right pilot port of the control valve 175R via the proportional valve 31AR, independently of the arm opening operation by the operator. That is, the controller 30 can open the arm 5 in response to the arm

opening operation by the operator or independently of the arm opening operation by the operator.

[0097] Further, with this configuration, even when the arm closing operation is performed by the operator, the controller 30 can reduce the pilot pressures applied to the pilot ports on the closing side of the control valves 175 (the left pilot port of the control valve 175L and the right pilot port of the control valve 175R) as necessary, and forcibly stop the closing operation of the arm 5. The same applies to a case where the opening operation of the arm 5 is forcibly stopped when the arm opening operation is performed by the operator.

[0098] Alternatively, even when the operator is performing the arm closing operation, the controller 30 may control the proportional valve 31AR as necessary to increase the pilot pressures applied to the pilot ports on the opening side of the control valves 175 (the right pilot port of the control valve 175L and the left pilot port of the control valve 175R) opposite to the pilot ports on the closing side of the control valves 175, and forcibly return the control valves 175 to the neutral positions, thereby forcibly stopping the closing operation of the arm 5. The same applies to a case where the opening operation of the arm 5 is forcibly stopped when the arm opening operation is performed by the operator.

[0099] Although description will be omitted with reference to FIGS. 4B to 4D below, the same applies to a case where the operation of the boom 4 is forcibly stopped when the boom raising operation or the boom lowering operation is performed by the operator, a case where the operation of the bucket 6 is forcibly stopped when the bucket closing operation or the bucket opening operation is performed by the operator, and a case where the slewing operation of the upper slewing body 3 is forcibly stopped when the slewing operation is performed by the operator. The same also applies to a case where the traveling operation of the lower traveling body 1 is forcibly stopped when the traveling operation is performed by the operator.

[0100] As illustrated in FIG. 4B, the right operation lever 26R is used to operate the boom 4. Specifically, the right operation lever 26R uses the hydraulic fluid discharged from the pilot pump 15 to apply pilot pressures corresponding to the operation in the front-rear direction to the pilot port of the control valve 174L and the pilot port of the control valve 174R. Specifically, when the right operation lever 26R is operated in the boom raising direction (rearward direction), the right operation lever 26R applies a pilot pressure corresponding to the operation amount to the right pilot port of the control valve 174L and the left pilot port of the control valve 174R. When the right operation lever 26R is operated in the boom lowering direction (forward direction), the right operation lever 26R applies pilot pressure corresponding to the operation amount to the right pilot port of the control valve 174R.

[0101] The operation sensor 29RA detects the operation content of the right operation lever 26R in the front-rear direction by the operator, and outputs the detected value to the controller 30.

[0102] The proportional valve 31BL operates in response to a control command (electric current command) output from the controller 30. The pilot pressure is adjusted by the hydraulic fluid introduced from the pilot pump 15 to the right pilot port of the control valve 174L and the left pilot port of the control valve 174R via the proportional valve

31BL. The proportional valves **31BR** operate in response to a control command (electric current command) output from the controller **30**.

[0103] The pilot pressure is adjusted by the hydraulic fluid introduced from the pilot pump **15** to the right pilot port of the control valve **174R** via the proportional valve **31BR**. The proportional valve **31BL** can adjust the pilot pressure so that the control valve **174L** and the control valve **174R** can be stopped at any positions. The proportional valve **31BR** can adjust the pilot pressure so that the control valve **174R** can be stopped at any positions.

[0104] With this configuration, the controller **30** can supply the hydraulic fluid discharged by the pilot pump **15** to the right pilot port of the control valve **31BL** and the left pilot port of the control valve **174L** via the proportional valves **174R** in response to the boom raising operation by the operator. The controller **30** can supply the hydraulic fluid discharged from the pilot pump **15** to the right pilot port of the control valve **174L** and the left pilot port of the control valve **174R** via the proportional valve **31BL**, independently of the boom raising operation by the operator. That is, the controller **30** can raise the boom **4** in response to the boom raising operation by the operator or independently of the boom raising operation by the operator.

[0105] Further, the controller **30** can supply the hydraulic fluid discharged from the pilot pump **15** to the right pilot port of the control valve **174R** via the proportional valve **31BR** in response to the boom lowering operation by the operator. Further, the controller **30** can supply the hydraulic fluid discharged from the pilot pump **15** to the right pilot port of the control valve **174R** via the proportional value **31BR**, independent of the boom lowering operation by the operator. That is, the controller **30** can lower the boom **4** in response to the boom lowering operation by the operator or independently of the boom lowering operation by the operator.

[0106] As illustrated in FIG. 4C, the right operation lever **26R** is also used to operate the bucket **6**. Specifically, the right operation lever **26R** uses the hydraulic fluid discharged from the pilot pump **15** to apply a pilot pressure corresponding to the operation in the right-left direction to the pilot ports of the control valve **173R**.

[0107] Specifically, when the right operation lever **26R** is operated in the bucket closing direction (left direction), the right operation lever **26R** applies a pilot pressure corresponding to the operation amount to the left pilot port of the control valve **173R**. When the right operation lever **26R** is operated in the bucket opening direction (right direction), the right operation lever **26R** applies a pilot pressure corresponding to the operation amount to the right pilot port of the control valve **173R**.

[0108] The operation sensor **29RB** detects the operation content of the right operation lever **26R** in the left-right direction by the operator, and outputs the detected value to the controller **30**.

[0109] The proportional valve **31CL** operates in response to a control command (electric current command) output from the controller **30**. The pilot pressure is adjusted by the hydraulic fluid introduced from the pilot pump **15** to the left pilot port of the control valve **173R** via the proportional valve **31CL**. The proportional valve **31CR** operates in response to a control command (electric current command) output from the controller **30**.

[0110] The pilot pressure is adjusted by the hydraulic fluid introduced from the pilot pump **15** to the right pilot port of

the control valve **173R** via the proportional valve **31CR**. The proportional valve **31CL** can adjust the pilot pressure so that the control valve **173R** can be stopped at any position. Similarly, the proportional valve **31CR** can adjust the pilot pressure so that the control valve **173R** can be stopped at any position.

[0111] With this configuration, the controller **30** can supply the hydraulic fluid discharged from the pilot pump **15** to the left pilot port of the control valve **173R** via the proportional value **31CL** in response to the bucket closing operation by the operator. The controller **30** can supply the hydraulic fluid discharged from the pilot pump **15** to the left pilot port of the control valve **173R** via the proportional value **31CL**, independent of the bucket closing operation by the operator. That is, the controller **30** can close the bucket **6** in response to the bucket closing operation by the operator or independently of the bucket closing operation by the operator.

[0112] Further, the controller **30** can supply the hydraulic fluid discharged by the pilot pump **15** to the right pilot port of the control valve **173R** via the proportional value **31CR** in response to the bucket opening operation by the operator. The controller **30** can supply the hydraulic fluid discharged from the pilot pump **15** to the right pilot port of the control valve **173R** via the proportional value **31CR**, independent of the bucket opening operation by the operator. That is, the controller **30** can open the bucket **6** in response to the bucket opening operation by the operator or independently of the bucket opening operation by the operator.

[0113] As illustrated in FIG. 4D, the left operation lever **26L** is also used to operate the slewing mechanism **2**. Specifically, the left operation lever **26L** uses the hydraulic fluid discharged from the pilot pump **15** to apply a pilot pressure corresponding to the operation in the left-right direction to the pilot port of the control valve **173L**.

[0114] Specifically, when the left operation lever **173L** is operated in the left slewing direction (left direction), the left operation lever **26L** applies a pilot pressure corresponding to the operation amount to the left pilot port of the control valve **SL**. When the left operation lever **26L** is operated in the right slewing direction (right direction), the left operation lever **26L** applies a pilot pressure corresponding to the operation amount to the right pilot port of the control valve **173L**.

[0115] The operation sensor **29LB** detects the content of the operation in the left-right direction performed on the left operation lever **26L** by the operator, and outputs the detected value to the controller **30**.

[0116] The proportional valve **31DL** operates in response to a control command (electric current command) output from the controller **30**. The pilot pressure is adjusted by the hydraulic fluid introduced from the pilot pump **15** to the left pilot port of the control valve **173L** via the proportional valve **31DL**. The proportional valve **31DR** operates in response to a control command (electric current command) output from the controller **30**.

[0117] The pilot pressure is adjusted by the hydraulic fluid introduced from the pilot pump **15** to the right pilot port of the control valve **173L** via the proportional valve **31DR**. The proportional valve **31DL** can adjust the pilot pressure so that the control valve **173L** can be stopped at any position. Similarly, the proportional valve **31DR** can adjust the pilot pressure so that the control valve **173L** can be stopped at any position.

[0118] With this configuration, the controller 30 can supply the hydraulic fluid discharged from the pilot pump 15 to the left pilot port of the control valve 173L via the proportional value 31DL in response to the left slewing operation by the operator. Further, the controller 30 can supply the hydraulic fluid discharged from the pilot pump 15 to the left pilot port of the control valve 173L via the proportional value 31DL, independent of the left slewing operation by the operator. That is, the controller 30 can cause the slewing mechanism 2 to slew to the left, in response to the left slewing operation by the operator or independently of the left slewing operation by the operator.

[0119] Further, the controller 30 can supply the hydraulic fluid discharged from the pilot pump 15 to the right pilot port of the control valve 173L via the proportional value 31DR in response to the right slewing operation by the operator. Further, the controller 30 can supply the hydraulic fluid discharged from the pilot pump 15 to the right pilot port of the control valve 173L via the proportional value 31DR, independent of the right slewing operation by the operator. That is, the controller 30 can turn the slewing mechanism 2 to the right in response to the right slewing operation by the operator or independently of the right slewing operation by the operator.

[0120] Next, a machine guidance function and a machine control function of the excavator 100 will be described with reference to FIG. 5. FIG. 5 is a block diagram illustrating an example of a configuration related to a machine guidance function and a machine control function of the excavator.

[0121] The controller 30 executes control of the excavator 100 related to a machine guidance function of guiding (leading) manual operation of the excavator 100 by the operator, for example.

[0122] The controller 30 transmits work information such as a distance between the target construction surface and the distal end portion of the attachment, that is, the work portion of the end attachment to the operator through the display device D1, the sound output device D2, or the like.

[0123] Specifically, the controller 30 acquires information from the boom angle sensor S1, the arm angle sensor S2, the bucket angle sensor S3, the body inclination sensor S4, the slewing state sensor S5, the space recognition device 81, the positioning device 84, the input device 83, and the like.

[0124] The data related to the target construction surface is stored in the internal memory, an external storage device connected to the controller 30, or the like, for example, based on a setting input through the input device 83 by the operator or by being downloaded from the outside (e.g., a predetermined management server).

[0125] The data related to the design surface is expressed by, for example, a reference coordinate system. The reference coordinate system is, for example, a world geodetic system. The world geodetic system is a three dimensional orthogonal XYZ coordinate system with the origin at the center of gravity of the earth, the X-axis in the direction of the intersection of the Greenwich meridian and the equator, the Y-axis in the direction of 90 degrees east longitude, and the z-axis in the direction of the north pole. For example, the operator may define any given point in the construction site as a reference point and set the design surface based on a relative positional relationship with the reference point through the input device 83.

[0126] The work portion of the bucket 6 is, for example, the claw tip of the bucket 6, the back surface of the bucket

6, or the like. Further, when a breaker is employed as the end attachment instead of the bucket 6, for example, the tip of the breaker corresponds to the work portion. Thus, the controller 30 can notify the operator of the work information through the display device D1, the sound output device D2, and the like, and can guide the operator to operate the excavator 100 through the operation device 26.

[0127] Further, the controller 30 executes control of the excavator 100 related to a machine control function of supporting manual operation of the excavator 100 by an operator or automatically or autonomously operating the excavator 100, for example. Specifically, the controller 30 is configured to acquire a target trajectory that is a trajectory followed by a position serving as a control reference (hereinafter, simply referred to as a “control reference”) set in a work portion or the like of the attachment.

[0128] In the case where there is a work target (e.g., the ground or the earth and sand on a bed of a dump truck described later) with which the end attachment can come into contact during work such as excavation work or rolling compaction work, a work portion of the end attachment (e.g., a claw tip or a back surface of the bucket 6) may be set as the control reference. In addition, in the case of an operation in which there is no work target with which the end attachment can come into contact, such as a boom raising and slewing operation, an earth and sand discharge operation, or a boom lowering and slewing operation, which will be described later, any given portion (e.g., the lower end portion or the claw tip of the bucket 6) that can define the position of the end attachment in the operation may be set as the control reference.

[0129] For example, the controller 30 derives the target trajectory based on the data related to the target construction surface. The controller 30 may derive the target trajectory based on the information on the terrain around the excavator 100 recognized by the space recognition device 81. The controller 30 may derive information on a previous trajectory of the work portion such as the claw tip of the bucket 6 from a previous output of a posture detection device temporarily stored in the internal volatile storage device, and derive the target trajectory based on the information. The controller 30 may derive the target trajectory based on the current position of a predetermined portion of the attachment and the data related to the target surface.

[0130] The posture detection device includes, for example, the boom angle sensor S1, the arm angle sensor S2, the bucket angle sensor S3, the blade angle sensor 96, and the like.

[0131] For example, when the operator manually performs excavation work, leveling work, or the like of the ground, the controller 30 automatically operates at least one of the boom 4, the arm 5, and the bucket 6 so that the target surface and a tip position of the bucket 6, specifically, a work portion such as the claw tip or the back surface of the bucket 6, are aligned with each other.

[0132] Specifically, when the operator performs an operation in the front-rear direction on the left operation lever 26L while operating (pressing) the switch SW, the controller 30 automatically operates at least one of the boom 4, the arm 5, and the bucket 6 in response to the operation so that the target surface and the tip position of the bucket 6 are aligned with each other. More specifically, as described above, the controller 30 controls the proportional valves 31 to automatically operate at least one of the boom 4, the arm 5, and

the bucket 6. Thus, the operator can cause the excavator 100 to perform excavation work, leveling work, or the like along the target surface only by operating the left operation lever 26L in the front-rear direction.

[0133] Further, for example, when a predetermined condition (hereinafter, referred to as an “excavation start condition”) is satisfied after the boom lowering and slewing operation of the excavator 100, the controller 30 may automatically perform the excavation operation in accordance with the operation of the attachment by the operator and move the bucket 6 in accordance with a predetermined target trajectory. The excavation start condition is a condition indicating the start of the excavation operation after the boom lowering and slewing operation of the excavator 100. For example, the excavation start condition may include a condition that “an operation related to the arm 5 has been performed (i.e., the left operation lever 26L has been operated in the front-rear direction) in a state where the bucket 6 is above the target surface”.

[0134] As described above, when a predetermined condition, that is, a condition corresponding to “operation on an operation target that has not been operated is started through a predetermined operation part (e.g., the operation device 26)” is satisfied, the controller 30 automatically causes the excavator 100 to perform, in accordance with the operation on the operation target, a predetermined operation to move a predetermined portion of the attachment in accordance with the target trajectory.

[0135] When an operation on the attachment is input, the controller 30 of the present embodiment detects whether the person is present around the excavator 100, and when a person is not present, the controller 30 performs a blade lowering operation of lowering the blade 95 to cause the blade 95 to be in contact with the ground, and prohibits the excavator 100 from traveling, based on the image or the like acquired by the space recognition device 81. The controller 30 may also automatically perform the excavation operation in accordance with the operation of the operator related to the attachment.

[0136] Therefore, in the present embodiment, the excavation start condition of the excavator 100 may include a “state where no person is detected around the excavator 100 and the blade 95 is in contact with the ground” in addition to “an operation related to the arm 5 being performed (i.e., the left operation lever 26L is operated in the front-rear direction) in a state where the bucket 6 is located above the target surface”.

[0137] Further, for example, when a predetermined condition (hereinafter, referred to as a “boom raising and slewing start condition”) is satisfied, the controller 30 automatically performs a raising operation of the boom 4 and the like in accordance with the slewing operation by the operator, and moves the bucket 6 along a predetermined target trajectory.

[0138] The boom raising and slewing start condition is a condition indicating the start of work for moving earth and sand or the like stored in the bucket 6 toward a dump truck parked at a predetermined position. For example, the boom raising and slewing start condition may include a condition that “the operation direction of the left operation lever 26L has been switched from the front-rear direction to the right-left direction in a state where the machine control function is enabled, that is, in a state where the switch SW is pressed”, as will be described later.

[0139] Further, for example, the boom raising and slewing start condition may include a condition that “the left operation lever 26L is operated in the left direction or in the right direction in a state where a predetermined switch (hereinafter, referred to as a “boom raising and slewing start switch”) provided at the distal end portion of the left operation lever 26L, which can be included in the input device 83, is pressed”. Further, for example, the boom raising and slewing start condition may include a condition that “the amount of earth and sand excavated by the attachment has become equal to or larger than a predetermined amount”.

[0140] Further, for example, the boom raising and slewing start condition may include “completion of excavation by the attachment by a predetermined distance or more”. In this case, the controller 30 can identify the amount of earth and sand excavated by the attachment, the excavation distance, and the like based on an image of the front of the upper slewing body 3 captured by a monocular camera or a stereo camera that can be included in the space recognition device 81, for example.

[0141] That is, the boom raising and slewing start condition is a condition for determining whether one operation of the excavator 100 such as the excavation operation is completed, for example. In addition, when the boom raising and slewing start condition includes a plurality of conditions as described above, the boom raising and slewing start condition may be satisfied when any one of the plurality of conditions included is satisfied, or the boom raising and slewing start condition may be satisfied when some of two or more of the plurality of conditions included, or all of the plurality of the conditions included are satisfied.

[0142] The same applies to an earth and sand discharge start condition, a boom lowering slewing start condition, and the like, which will be described later. Specifically, when the operator operates the left operation lever 26L in the left direction or in the right direction, the controller 30 automatically operates at least the boom 4 of the upper slewing body 3 and the attachment in response to the operation so that the target trajectory and a portion (e.g., the lower end portion of the bucket 6) serving as a control reference of the bucket 6 are aligned with each other.

[0143] More specifically, the controller 30 controls the proportional valves 31 to automatically operate the upper slewing body 3, the boom 4, and the like as described above. Thus, the operator can cause the excavator 100 to perform the boom raising and slewing operation for moving the earth and sand and the like stored in the bucket 6 to the dump truck by simply operating the left operation lever 26L in the left-right direction.

[0144] Further, for example, when a predetermined condition (hereinafter, referred to as an “earth and sand discharge start condition”) is satisfied, the controller 30 automatically performs the opening operation of the arm 5 and the like in accordance with the opening operation of the bucket 6, and discharges the earth and sand and the like stored in the bucket 6 toward the dump truck. The earth and sand discharge start condition is a condition indicating the start of work for discharging the earth and sand and the like stored in the bucket 6 to the dump truck.

[0145] For example, the earth and sand discharge start condition may include a condition that “the state where the left operation lever 26L is operated in the left-right direction is switched to the state where the right operation lever 26R

is operated in the left-right direction (specifically, the left direction corresponding to the opening operation of the bucket 6) in a state where the machine control function is enabled, that is, in a state where the switch SW is pressed”, as will be described later.

[0146] Further, for example, the earth and sand discharge start condition may include a condition that “the right operation lever 26R is operated in the left direction (the operation of closing the bucket 6) or in the right direction (the operation of opening the bucket 6) in a state where a predetermined switch (hereinafter, referred to as “an earth discharge start switch”) provided at the tip portion of the right operation lever 26R, which may be included in the input device 83, is pressed”.

[0147] Further, for example, the earth and sand discharge start condition may include a condition that “the bucket 6 has reached a predetermined point (e.g., an end point of the target trajectory) above the dump truck”. In this case, the “predetermined point (the end point of the target trajectory)” in the earth and sand discharge start condition may be changed every time the earth and sand discharge is performed. Specifically, when the operator operates the right operation lever 26R in the right direction, the controller 30 causes the bucket 6 to open and causes the arm 5 to open in response to the operation so that the earth and sand and the like in the bucket 6 are discharged to a predetermined target position on the bed of the dump truck.

[0148] More specifically, the controller 30 controls the proportional valves 31 to automatically operate the arm 5, the bucket 6, and the like as described above. Thus, the operator can discharge the earth and sand and the like stored in the bucket 6 onto the bed of the dump truck only by operating the right operation lever 26R in the left-right direction (specifically, in the right direction).

[0149] Further, for example, when a predetermined condition (hereinafter, referred to as a “boom lowering slewing start condition”) is satisfied, the controller 30 automatically performs the lowering operation of the boom 4 and the like in accordance with the slewing operation by the operator, and moves the bucket 6 in accordance with a predetermined target trajectory.

[0150] The boom lowering slewing start condition is a condition indicating the start of work for slewably moving the attachment to the original position for performing excavation work or the like after discharging the earth and sand or the like in the bucket 6 onto the bed of the dump truck. For example, the boom lowering slewing start condition may include a condition that “the right operation lever 26R is switched from a state of being operated in the left-right direction (specifically, the right direction) to a state of the left operation lever 26L being operated in the left-right direction” as will be described later.

[0151] Further, for example, the boom lowering slewing start condition may include a condition that “the left operation lever 26L is operated in the left direction or in the right direction in a state where a predetermined switch (hereinafter, referred to as a “boom lowering slewing start switch”) which can be included in the input device 83 and is provided at the tip portion of the left operation lever 26L is pressed”. Further, for example, the boom lowering slewing start condition may include a condition that “there is no earth and sand falling from the bucket 6 onto the bed of the dump truck”.

[0152] In this case, the controller 30 can identify the amount of earth and sand or the like in the bucket 6 based on, for example, an image of the front of the upper slewing body 3 captured by a monocular camera or a stereo camera which can be included in the space recognition device 81. Specifically, when the operator operates the left operation lever 26L in the left direction or in the right direction, the controller 30 automatically operates at least the boom 4 of the upper slewing body 3 and the attachment in response to the operation so that the target trajectory and the portion serving as the control reference of the bucket 6 are aligned with each other.

[0153] More specifically, the controller 30 controls the proportional valves 31 to automatically operate the upper slewing body 3, the boom 4, and the like as described above. Thus, the operator can cause the excavator 100 to perform the boom lowering and slewing operation of moving the attachment to the original position for the excavation work or the like after discharging the earth and sand or the like stored in the bucket 6 to the bed of the dump truck by only operating the left operation lever 26L in the lateral direction.

[0154] Further, for example, when a predetermined condition (hereinafter, referred to as a “bucket leveling operation start condition”) is satisfied before the boom lowering and slewing operation of the excavator 100, the controller 30 may automatically perform an operation (hereinafter, referred to as a “bucket leveling operation”) for leveling earth and sand or the like loaded on the bed of the dump truck in response to the operation of the attachment by the operator and move the bucket 6 in accordance with a predetermined target trajectory.

[0155] The bucket leveling operation start condition is a condition indicating the start of the bucket leveling operation after the earth and sand or the like in the bucket 6 are discharged onto the bed of the dump truck. For example, the bucket leveling operation start condition may include a condition that “there are no earth and sand falling from the bucket 6 onto the bed of the dump truck”.

[0156] Further, for example, the bucket leveling operation start condition may include a condition that an “operation related to the arm 5 is performed (i.e., the left operation lever 26L is operated in the front-rear direction) in a state where the bucket 6 is present above the bed of the dump truck”. In this case, the controller 30 may generate the target trajectory based on the shape of the bed of the dump truck, which is defined in advance and stored in an internal or external communicating nonvolatile storage device.

[0157] Further, when an operation instructing normal traveling is input after the operation on the attachment is ended, the controller 30 detects whether a person is present around the excavator 100, based on the image or the like acquired by the space recognition device 81, and when the person is not present, a blade raising operation of raising the blade 95 is performed.

[0158] In other words, when an operation instructing normal traveling is input after the excavation operation is ended and no person is detected around the excavator 100, the controller 30 raises the blade 95 to cause the excavator 100 to travel.

[0159] The controller 30 of the present embodiment may automatically perform an operation (hereinafter, referred to as a “normal traveling operation”) of moving the lower traveling body to the target position along a predetermined target trajectory. The controller 30 may also use the condi-

tion that the blade 95 is separated from the ground and the condition that the bucket 6 has risen to a predetermined height from the ground as the conditions for starting normal traveling. Further, the controller 30 may automatically perform the operation of raising the blade 95 and the operation of raising the attachment.

[0160] Further, when an operation for instructing the leveling travel is input after the operation on the attachment is ended, the controller 30 detects whether a person is present around the excavator 100 based on the image or the like acquired by the space recognition device 81, and performs the blade lowering operation of lowering the blade 95 when the person is not present.

[0161] In other words, when the operation for instructing the leveling travel is input and no person is detected around the excavator 100, the controller 30 lowers the blade 95 to cause the excavator 100 to travel. Further, the controller 30 may automatically perform an operation (hereinafter, referred to as a “leveling travel operation”) of moving the lower traveling body along a target trajectory set in a predetermined region in order to level the ground in the predetermined region by the blade. The controller 30 may set the leveling travel start condition to be that the blade 95 is in contact with the ground and that the bucket 6 has risen to a predetermined height from the ground. Further, the operation of lowering the blade 95 and the operation of raising the attachment may be automatically performed. The controller 30 controls the blade 95 and the attachment based on at least one of the detection values of the space recognition device 81 and the posture detection device.

[0162] The following description will be given on the assumption that the machine control function is enabled when the left operation lever 26L and the right operation lever 26R are operated in a state where the switch SW is pressed.

[0163] Although not illustrated in FIG. 5, the excavator 100 may include a communication device for transmitting and receiving information to and from an external device via a network or the like, and information received by the communication device may be input to the controller 30. The controller 30 may transmit the information acquired by the controller 30 to an external device via the communication device. The external device may be a support device for supporting work by the excavator 100, a management device for managing a work portion of the excavator 100, or the like.

[0164] Next, the operation of the excavator 100 according to the present embodiment will be described with reference to FIG. 6. FIG. 6 is a flowchart illustrating the operation of the excavator.

[0165] In the excavator 100 according to the present embodiment, the controller 30 receives an input of an attachment operation command instructing an operation of the attachment (step S601). In other words, the excavator 100 receives an operation of the attachment by the operator.

[0166] Subsequently, the controller 30 determines whether a person is detected around the excavator 100 (step S602). Specifically, for example, the controller 30 may determine whether a person is detected around the excavator 100 using image data around the excavator 100 acquired by the space recognition device 81.

[0167] When a person is detected in step S602, the controller 30 proceeds to step S609 described later.

[0168] When no person is detected in step S602, the controller 30 performs a blade lowering operation to lower the blade 95 (step S603). In particular, the controller 30 performs control to extend the blade cylinder 95A.

[0169] Subsequently, the controller 30 determines whether both ends of the tip portion 95a of the blade 95 are in contact with the ground (step S604). The controller 30 may determine whether both ends of the tip portion 95a are in contact with the ground based on, for example, a signal output from the blade angle sensor 96, or may determine whether both ends of the tip portion 95a are in contact with the ground based on an image captured by the space recognition device 81.

[0170] In step S604, when both ends of the tip portion 95a of the blade 95 are not in contact with the ground, the process proceeds to step S609 described below.

[0171] In step S604, when both ends of the tip portion 95a of the blade 95 are in contact with the ground, the controller 30 prohibits the excavator 100 from traveling while the attachment is operating (step S605). Specifically, the controller 30 may disable the input by the travel lever.

[0172] Subsequently, after the operation on the attachment is ended, the controller 30 receives a travel command for instructing the excavator to travel (step S606). Specifically, the controller 30 receives an input from the travel lever in a state where there is no input of an operation command for instructing an operation of the attachment (no operation on the operation lever).

[0173] Subsequently, the controller 30 determines whether a person is detected around the excavator 100 (step S607). When no person is detected in step S607, the controller 30 performs a blade raising operation to raise the blade 95 (step S608). Specifically, the controller 30 performs control to contract the blade cylinder 95A.

[0174] When a person is detected in step S607, the controller 30 stops the operation of the excavator 100 (step S609).

[0175] At this time, the controller 30 may output an alarm or the like by the display device D1 or the sound output device D2.

[0176] In the present embodiment, when an operation on the attachment is received, the blade 95 is lowered and brought into contact with the ground. Therefore, in the present embodiment, while the attachment is being operated, the tip portion 95a of the blade 95 is in contact with the ground, and it is possible to reduce shaking or the like of the body during work.

[0177] In the present embodiment, when the attachment is not operated and the operation for instructing the traveling is performed, the blade 95 is raised. Therefore, in the present embodiment, for example, when the excavator 100 performs excavation work, the operator does not need to perform an operation of lowering the blade 95 every time the operator performs an operation of the attachment. In the present embodiment, when the excavator 100 travels after finishing the excavation work, the operator does not need to perform the operation of raising the blade 95.

[0178] Therefore, according to the present embodiment, the operation by the operator can be simplified, and the operability can be improved.

[0179] In particular, in the present embodiment, the blade 95 automatically lowers during excavation by the excavator 100 and automatically rises during traveling of the excavator 100. Therefore, the present embodiment is effective in a case

where the excavation and the traveling are repeated during work such as work of digging a trench, and is capable of improving the work efficiency of the excavator 100.

[0180] Further, in the present embodiment, the blade 95 is operated only when no person is detected around the excavator 100. Therefore, in the present embodiment, for example, when a person enters a range that cannot be visually observed from the operator seated in the cabin 10, the person entered in the range can be detected, and safety can be improved.

[0181] In the present embodiment, for example, when deep digging work is performed in which the bucket 6 is located below the lower traveling body 1, the lower traveling body 1 may be slewed so that the blade 95 is located on the rear side when the direction in which the cabin 10 faces is the front side.

[0182] In this case, for example, the controller 30 may calculate the position of the bucket 6 based on the input of an operation command of the attachment, and when the bucket 6 is located below the lower traveling body 1, the controller 30 may instruct the operator to slew the lower traveling body 1 so that the blade 95 is located on the rear side. Specifically, the controller 30 may cause the display device D1 to display a message for instructing the operator to slew the lower traveling body 1 so that the blade 95 is located on the rear side, or may cause the sound output device D2 to output a sound guidance.

[0183] In the present embodiment, as described above, when the deep digging work is performed, the blade 95 is located on the rear side, and thus it is possible to prevent the blade 95 and the boom 4 from coming into contact with each other.

[0184] In the present embodiment, when the blade 95 is raised in response to reception of the travel instruction, the blade 95 may be raised to a height at which the tip of the blade 95 is not in contact with the ground, and the ground leveling work may be performed. Whether to perform the ground leveling work in response to reception of the travel instruction may be set in advance by the operator.

[0185] Next, a case where the blade 95 is lowered in the excavator 100 according to the present embodiment will be described with reference to FIGS. 7A and 7B. FIGS. 7A and 7B include views illustrating a situation when the blade is lowered. FIG. 7A is a view illustrating a state where the tip portion 95a of the blade 95 is in contact with the ground, and FIG. 7B is a view illustrating a state during the deep excavation work.

[0186] In the present embodiment, as illustrated in FIG. 7A, the blade 95 is lowered until both ends of the tip portion 95a of the blade 95 are in contact with the ground G1 which is the traveling surface by the excavator 100. In the present embodiment, when it is detected that both ends of the tip portion 95a of the blade 95 are in contact with the ground G1, the operation of the attachment is started.

[0187] Therefore, according to this embodiment, the ground contact area between the body of the excavator 100 and the ground G1 is increased even when, for example, a groove G2 is formed in the ground G1. Therefore, according to this embodiment, excavation work can be performed with the body in a stable condition, and work efficiency can be improved.

[0188] Further, in the excavator 100, for example, when the deep digging work in which the bucket 6 is located below the lower traveling body 1 is performed, as indicated

by an oval 97 of FIG. 7B, the blade 95 and the boom cylinder 7 may come into contact with each other.

[0189] In the excavator 100 according to the present embodiment, when performing the deep digging work, the lower traveling body 1 is slewed so that the blade 95 is in a rearward position, and thus it is possible to avoid the blade 95 and the boom cylinder 7 from coming into contact with each other.

[0190] In the present embodiment, for example, the distance between the boom cylinder 7 and the blade 95 may be calculated based on the image captured by the space recognition device 81F, and when the distance is equal to or less than a certain value, a warning may be given to avoid the contact.

[0191] The distance between the boom cylinder 7 and the blade 95 may be calculated from, for example, the detection value of the blade angle sensor 96 and the detection value of the boom angle sensor S1. The distance between the boom cylinder 7 and the blade 95 may be calculated based on, for example, image data captured by an imaging device provided around the excavator 100. The imaging device provided around the excavator 100 may be, for example, a fixed-point camera or the like installed at a work portion, or may be an imaging device provided in a flying object flying around the excavator 100.

[0192] In the present embodiment, the controller 30 of the excavator 100 executes the processes illustrated in FIG. 6, but the present disclosure is not limited thereto.

[0193] The controller 30 of the present embodiment may be, for example, a control part provided in an external device outside the excavator 100.

[0194] Specifically, the controller 30 may be implemented by a control part of a support device for remotely controlling the excavator 100 or a management device that manages the excavator 100.

[0195] In this case, the support device or the management device transmits an operation command of the attachment or the travel command to the excavator 100 in response to an operation of the excavator 100 by the remote operator. In the excavator 100, when receiving these commands, the controller 30 causes the excavator 100 to perform an operation according to the corresponding command.

[0196] Therefore, according to the present embodiment, even when the excavator 100 is remotely operated, the excavator 100 can be caused to automatically raise and lower the blade 95, and operability can be improved.

[0197] The scope of the present disclosure is not limited to the technical matters described in the above-described embodiment, and design changes and the like within the scope not departing from the gist of the present disclosure are also included in the scope of the present disclosure.

What is claimed is:

1. A control device for an excavator, the excavator including an upper slewing body, an attachment provided on the upper slewing body, a slewing mechanism, a lower traveling body, and a blade provided on the lower traveling body, the control device comprising:

circuitry configured to receive an input of an operation command for instructing an operation of the attachment, determine whether a person is detected around the excavator, and cause the excavator to perform an operation of lowering the blade when the person is not detected.

2. The control device according to claim 1, wherein the circuitry is configured to prohibit the excavator from traveling while performing the operation of lowering the blade.

3. The control device according to claim 1, wherein the circuitry is configured to receive an input of a travel command for instructing the excavator to travel, and cause the excavator to perform an operation of raising the blade.

4. The control device according to claim 3, wherein the circuitry is configured to determine whether both ends of a tip portion of the blade are in contact with a ground after the operation of lowering the blade is performed, and output an alarm when the both ends are not in contact with the ground.

5. The control device for an excavator according to claim 4, wherein the circuitry is configured to output an alarm for avoiding contact between the blade and the attachment in accordance with a distance between the blade and the attachment when the operation indicated by the operation command is an operation by which an end attachment included in the attachment is located below the lower traveling body.

6. The control device for an excavator according to claim 1, wherein the circuitry is configured to receive an input of a travel command for instructing the excavator to travel, and cause the attachment to perform an operation of raising the attachment.

* * * * *